

## Education

### Indiana University

B.S. in Computer Science & B.S. in Information Systems

Courses taken: Software Engineering for Information Systems, Distributed Systems, Database Concepts, Applied Algorithms, Data Structures, Mobile App Development (Android), Object-Oriented Software Methods, Systems Programming with C & Linux, Introduction to Data Analysis & Mining, Introduction to Artificial Intelligence

## Skills

Python, Java, React, Reach Native, Django, SpringBoot, JavaScript, Node.JS, Express.JS, C, Linux, C#, Kotlin, Agile, Swift, HTML, PostgreSQL, SQL, NoSQL, RestAPI, CSS, R, Git, Docker, Tableau, MongoDB, CI/CD

## Experience

### Machine Learning Intern

May 2026 – September 2026

Jakarta, Indonesia

- Built an end-to-end pricing recommendation pipeline in Python for near-expiry inventory, covering data generation, feature engineering, model training, and inference
- Trained an XGBoost regression model to predict markdown percentages from product category, days to expiry, stock level, sales velocity, and calendar features, evaluated with MAE and  $R^2$  on a held-out test set
- Designed the schema and a synthetic dataset (3,000 records) to validate the pipeline ahead of real company data, with a documented path to upgrade to a sell-through probability and price optimization approach

### Research Assistant

February 2025 – December 2025

Indiana University

Bloomington, IN, USA

- Engineered scalable Python pipelines integrating OCR, text cleaning, and OpenAI APIs to process **2,900+** scam-related images and videos, extracting and classifying content using **Mistral-7B and Llama-2** via Ollama and LM Studio to enhance dataset usability for NLP research
- Trained a **BERT-based** multi-label classification model in **PyTorch** to automate detection of psychological pressure tactics in scam posts, implementing end-to-end data preprocessing, model training, and performance evaluation workflows
- Led collaborative development of standardized annotation framework, conducting comprehensive literature review and analyzing **500+ posts** to establish robust labeling guidelines for pressure tactics and audience targeting

### Mobile App Development Associate Instructor

August 2024 – December 2024

Indiana University

Bloomington, IN, USA

- Mentored **30 students** in Android development using **Kotlin and Android Studio**, providing technical guidance on mobile app architecture, performance optimization, and debugging methodologies
- Conducted live code walkthroughs demonstrating systematic problem-solving approaches and best practices for resolving common development challenges

## Projects

### MoodMosaic

<https://play.google.com/store/apps/details?id=com.moodmosaic&hl=en>

- Architected and deployed production **React Native** mental wellness application serving real users on Google Play Store, implementing **Firebase** backend for cloud-based mood tracking with real-time data synchronization
- Integrated **Cloudflare R2** for scalable media storage leveraging **S3-compatible API** for cost-effective asset management
- Designed interactive data visualization features enabling users to identify emotional patterns and track mental wellness trends over time

### RateMyUniversity

<https://ratemyuniversity.io>

- Developed full-stack university review platform supporting authenticated and anonymous users, implementing secure **JWT-based authentication** and **PostgreSQL** for user and session management
- Built responsive **React** frontend with interactive UI components using hooks and state management patterns to enhance user engagement
- Engineered access control and rate-limiting middleware for production-grade APIs supporting authenticated and anonymous users